

Problem Set 6

Due on Monday August 23 at 11:59 PM (if you want feedback)

1. Let $A \in M^{8 \times 5}$ be given.

- (a) What is the largest possible value of $\dim(\mathcal{N}(A))$?
- (b) What is the smallest possible value of $\dim(\mathcal{N}(A))$?

2. Compute $\det(A)$ if

- (a) $A = \begin{pmatrix} -3 & 5 \\ 2 & 2 \end{pmatrix}$
- (b) $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -2 & 4 \\ 1 & 1 & 1 \end{pmatrix}$

3. Find the eigenvalues of A if

- (a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$
- (b) $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -5 & 0 \\ 4 & 6 & \sqrt{2} \end{pmatrix}$

4. Let x, y, z be given real numbers and put

$$A = \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{pmatrix}$$

Show that A is invertible if and only if x, y , and z are distinct.

5. Let $A \in M^{2 \times 4}$ and assume that the null space of A is given by

$$\mathcal{N}(A) = \{x \in V^4 : x_1 + x_3 = 0, x_2 - x_4 = 0\}.$$

What can you conclude about the rank of A ?

6. Let A be an $N \times N$ matrix, λ be a scalar, and $x \in V^N$ with $x \neq 0$. Assume that $x \in \mathcal{N}((A - \lambda I)^2)$.

- (a) Can we conclude that x is an eigenvector of A ?
- (b) Can we conclude that λ is an eigenvalue of A ?